

Machine Learning

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Introduction

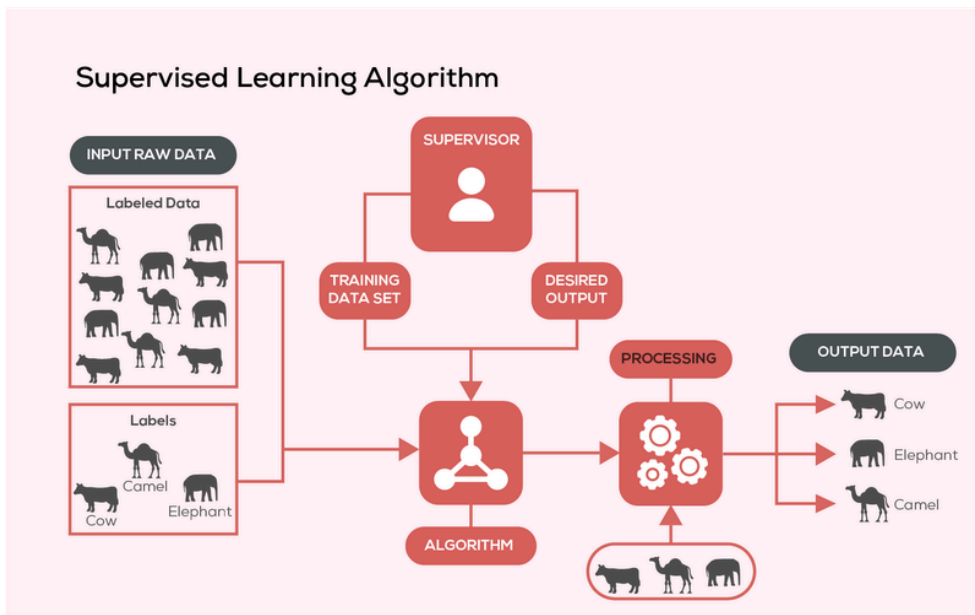
Machine Learning (ML) is a part of Artificial Intelligence that allows computers to learn from data and make decisions without being directly programmed.

Instead of following fixed rules, the system learns patterns from data and improves over time.

Machine learning is used in many real-life applications like recommendation systems, spam detection, and face recognition

Types

Supervised Learning

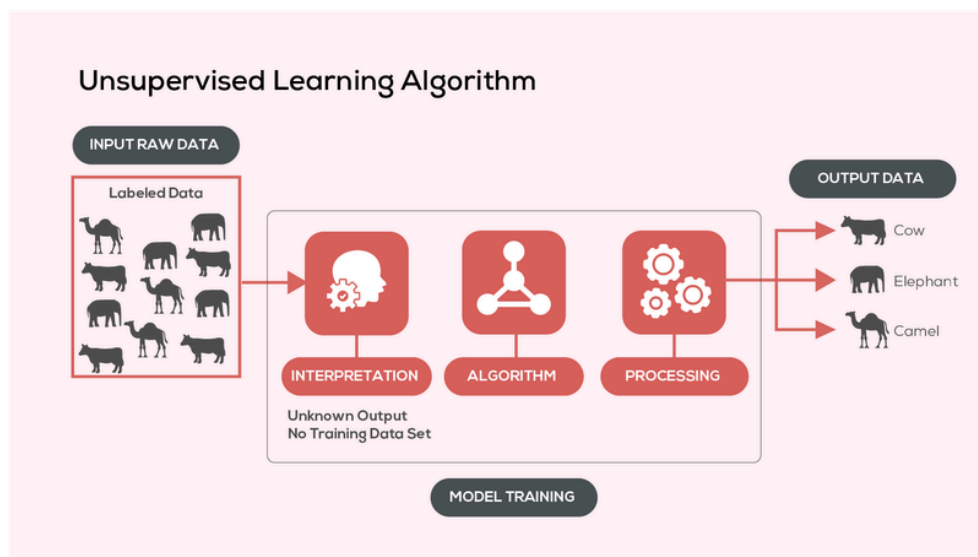


Supervised learning is a type of machine learning where a model learns from labelled data, meaning each input has a correct output. The model compares its predictions with actual results and improves over time to increase accuracy.

Types of Supervised Learning

- Classification: Where the output is a categorical variable (e.g., spam vs. non-spam emails, yes vs. no).
- Regression: Where the output is a continuous variable (e.g., predicting house prices, stock prices).

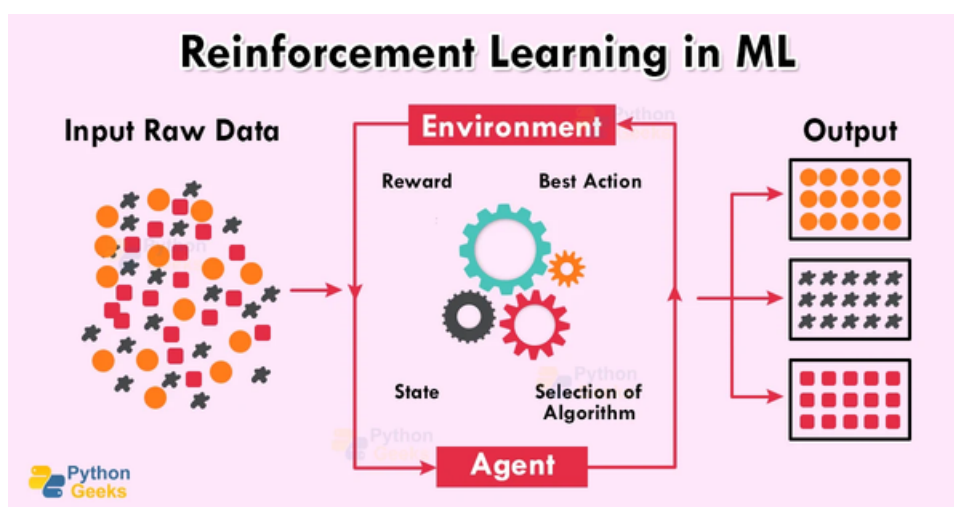
Unsupervised Learning



Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.

- It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.
- Helps identify hidden patterns in data
- Useful for grouping, compression and anomaly detection

Reinforcement Learning



Reinforcement Learning (RL) is a branch of machine learning that focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards. RL allows machines to learn by interacting with an environment and receiving feedback based on their actions. This feedback comes in the form of rewards or penalties.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression, which predicts continuous values it predicts the probability that an input belongs to a specific class.

- It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1.
- It uses sigmoid function to convert inputs into a probability value between 0 and 1.

-Example:

- Will a passenger survive or not (Titanic)

Types

Logistic Regression can be divided into three types based on the kind of output we are trying to predict.

Binomial Logistic Regression

This is used when there are only two possible outcomes.

For example, predicting:

- Yes or No
- Pass or Fail

This is the most common type and is mainly used for simple classification problems.

Multinomial Logistic Regression

This is used when there are more than two categories, and there is no specific order between them.

For example:

- Classifying animals as cat, dog, or sheep

Here, the model just decides which category the input belongs to.

Ordinal Logistic Regression

This is used when there are multiple categories with a clear order or ranking.

For example:

- Low, Medium, High

In this case, the order of the categories is important, and the model takes that into account while making predictions.

Sigmoid Function

The sigmoid function is an important part of logistic regression. It is used to convert the model's output into a probability value between 0 and 1.

This function takes any real number as input and transforms it into a value between 0 and 1, creating an S-shaped curve, known as the sigmoid or logistic curve. Since probabilities always lie between 0 and 1, this function is perfect for classification problems.

In logistic regression, we usually use a threshold value of 0.5 to make predictions:

- If the output is greater than or equal to 0.5, it is classified as Class 1
- If the output is less than 0.5, it is classified as Class 0

This helps in converting continuous values into clear class labels, making the model useful for decision-making

